Engineering Mechanics 1 · Unit_03 · UNIT 3

DRAFT — UE26ME142A · Unit_03 · UNIT 3 · UE25CV141A Unit 3 Class 53 AVSummary

Standalone video lesson · authored study notes with source citations

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· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

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M1

UE25CV141A Unit 3 Class 53 AVSummary

1 UE25CV141A Unit 3 Class 53 AVSummary

Study focus: UE25CV141A_Unit 3_Class 53_AVSummary. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000134 · EV0000135

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

My name is Dr Sv Vingitish from Civil Engineering under the Department of Science and Humanities, PS University.

Key details

- This is class number fifty-three.
- We are going to be starting the new topic in unit three under distributed forces, that is Teams internal effects.
- Written by Miriam and Craig.
- It's the eighth edition and SI Mosh.
- Chapter five, Beams Internal Effect is the.
- Determinate and indeterminate based on the type of support and the number of supports that is going to be acting on the beam.
- If it is equal to the number of equations of equilibrium or less than the number of equations of equilibrium, it is determined B.
- If the number of supports, external supports, are more than the available number of equations of equilibrium.
- Then they are treated as indeterminate beams.
- However, the indeterminate beams can be determined or can be the support reactions can be determined using other equations.
- The external effect or the support reactions which we have found or which we know how to find, but now we are trying to find what are the internal forces that is going to be acting in the beam because of the application of the external forces and as well as the external support reactions.
- They are shear force, bending movement, and torsion.

- Simply called a shear bending untorsion.
- In addition to the.
- Tension and compression that the beam is going to take.
- A beam has to resist all these three forces that is shear bending and torsion.
- The first one indicates the shear force.
- There could be forces acting whatever is indicated here.
- This will make that section two separate out or it gets sheared off.
- If the forces are acting like this, then it is going to get sheared off or the beam is going to get separated.
- That is what is called as shear force.
- Next coming to bending moment, if we apply movement like this, okay.
- Then the beam is going to bend because of this bending, it is called as bending movement.
- Earlier we used to find moment because of all the forces that is acting on a particular beam.
- But the bending moment is we are going to find out the moment acting only to the left side of the section or to the right side of the section.
- Moment at any particular point is equal to zero as the one of the equations of equilibrium.
- That is going to be applied when we take entire beam as one, including the support reactions, the applied forces, everything put together, what is the moment that is acting at a particular point should be equal to zero.
- But whereas bending moment, we are taking only the left side of the portion and whatever is the load that is acting in the left side of the portion, because of that whatever is the moment that is going to be coming, that is treated as bending moment.
- Or instead of taking left side, we can also take to the right side.
- For a section, whatever is the force that is acting to the right side of the section on the beam, if we take the movement of all those forces at that section, then whatever value that we are going to get will be called as the bending moment.

Example or application from the lecture

- You can see the different diagram that is given.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000134 · EV0000135 · EV0000136 · +1 source units in pack.json

Core explanation · 05:00–09:59

This beam is subjected to two movements.

Key details

- One is clockwise, the other one is anti-clockwise, at the two ends.
- Electrical wire between the two cables, something like this, it is going to bend.
- Just like your half moon.
- The top zone or the top fiber, the extreme top fiber would undergo a compression and it will be under reduction in the length and it will be under compression.
- Whereas the bottom one, the extreme bottom layer, if you take, it will go, it is going to elongate and that would be under tension.

- Is called as a neutral axis.
- The zone above the neutral axis or the neutral layer is under compression, and that is called as a compressive zone, whereas the zone which is below the neutral axis or the neutral layer is called as a tensile zone.
- Next, we have this that is the torsion.
- If the movement that is acting because of the application of the load along the axis of the beam, that is the longitudinal axis of the beam, that movement is referred to as torsion.
- The internal forces that is going to be acting are the combination of all these three, which is indicated here.
- There is bending moment M as well as there is torsion P torsional moment T acting along the longitudinal axis of the beam.
- Unlike the external effect where we used to have the supports, replace the support by its reaction, determine those reactions by using the equations of the curve.
- Here, when we have to find out the shear force of bending moment.
- We have to use a different sign convention.
- This is the sign convention that needs to be followed for shear.
- If there is a point load acting upwards, then we will take it as positive.
- If it is acting downwards, we will take it as negative.
- If it is to the right side of the section and if it is acting downwards,.
- If it is acting downwards, then we will take it as positive.
- If it is to the left side of the section, and if it is acting upwards, then we will take it as positive.
- Vice versa to this has to be taken as negative.
- If I take a section here and if I take the portion of the beam acting to the right side of the section, if it is anti-clockwise, that moment whatever we are going to act, okay, will bear make the beam bend like a.
- Electrical wire between the two poles, half of it.
- Half of the electrical wire between the two poles.
- If you take only one pole and half of the electrical wire, it is going to bend like this, bend like this, bend like this.
- If the bending moment causes the beam to sag, then we take it as positive.
- You can see that this is like half of the.

Example or application from the lecture

- Then I will consider as I have told you earlier either to the left side or to the right side.
- If there is a point lot that is acting downwards like this, what is indicated, we consider that as positive.
- Whereas if it is opposite to this, that is acting upwards, then we will consider it as negative.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000138 · EV0000139 · EV0000140 · +1 source units in pack.json

Core explanation · 10:00–14:59

Now next coming to this section, if you take a section and if you consider the left side portion of the beam.

Key details

- If the movement is clockwise, it is going to make the beam bend like a sagging, that is like a half moon, okay, or an electrical cable between the two poles.
- If it is on the opposite direction, that is if the movement here, if this movement would have been in the counterclockwise direction, then the beam would have been like this, which is similar to your half rainbow.
- You have cleared with the sign convention what needs to be used for shear force and many movement.
- However, we had three internal effects, but we'll be concentrating only on the shear force and many movement because torsional is not going to be coming here, but you should know what are the three forces that is acting.
- To understand what we need to do in this section is we need to draw the shear force diagram separately across the length of the beam.
- At any point along the length of the beam, if I have the diagram, I can find out what are the values of the shear force and bending movement.
- The design of the beam will be based on the values of the shear force and bending movement.
- The bending movement will actually talk about the longitudinal reinforcement that we need to be provided in a reinforcement concrete beam.
- Whereas the shear force will determine what are the stir-ups that is to be provided in the beam for designing that beam or to carry the load that is coming on that.
- I've chosen a problem 5 r 125.
- The statement of the problem goes like this.
- Determine the shear force and bending movement distribution.
- Distribution means across the length of the beam, what is the distribution of the shear force and bending moment.
- Produced in the beam by the concentrated load.
- And they have also asked you what are the values of the shear force and bending moment at x is equal to L by 2.
- This is what is asked.
- The other end of the beam is free, there is no support, so this is a cantilever beam.
- And there is a point load that is acting at a length of 2l pi 3 from A.
- That is from A to this point where the point load is acting, there is no load.
- And then we can get an equation for shear force and minimum diagram and we can substitute the value.
- of the point from A, whatever is the length, then we can get the shear force amenement diagram or shear force amendment at that point.
- And we can also get one more equation.
- So there will be two equations for this because there is only one point load.
- And substituting the value for in that equation, we will get the value of the sheer monument diagram at a particular point.

- This is a solution for drawing the bending mode and cheap for diagram.
- First thing that we need to do is we have to draw a free body diagram.
- Free body diagram is free of all the supports.
- The supports have been removed and replaced it by the respective reactions.
- Then we have the point load, distance is also mentioned, the beam is mentioned with respect to a horizontal line because the depth of the beam is not going to affect the shear force embedding moment.
- Now we need to first find out the.
- Support reactions for which we will use the equations of equilibrium.
- This is very well known to all of you.
- You have already solved several problems on external effects of the beam.

Example or application from the lecture

- So let's take a simple example to understand how do we draw the shear force and diagram.
- Now, if you read the diagram that is given, this is a cantilever beam because you have a fixed support at A.
- To draw the shear force unbending moment diagram, we should consider a section in the portion where there is no variation in the load.

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EV0000142 · EV0000143 · EV0000144 · +1 source units in pack.json

Core explanation · 15:00–19:59

Horizontal reaction is zero, vertical reaction is P.

Key details

- Moment we will get as 2PL by 3, which will be acting in the counterclockwise direction.
- Diagram or the bending moment.
- I will take a section here that is X.
- The distance from A to this section is considered as X itself.
- Section the beam or to the right side, if you take it to the left side, what we get for shear forces we have VA which is equal to P, that is what we have got it from here.
- Then if I take the movement MA, it is acting in the counterclockwise direction.
- So substituting the value of MAR and substituting the value of V A, so it becomes P into X minus because this MA, we know that it is Augging, so it is negative 2 P L by 3.
- This is the equation for shear force that is VA equals P, and the bending moment equation is P x.
- Minus 2PL by 3 is the equation for many limited.
- We also have this point load P.
- In this zone, that is from the point where the point load is acting to the end of the beam, shear force is zero.
- Next to take the bending moment, again we will take the moment, so we will get VA into hex distance.

- You have the moment MA which will be acting or making the beam to bend like a hogging, so it is negative, whereas VA is sagging, make the beam to bend like sagging, so it is positive.
- Using this, we can draw the shear force and bending movement diagram.
- But however, in the problem they have asked you to find out.
- The value of the shear force admitting moment at a point where x is equal to L by 2.
- Simplifying that we will get the value for the shear force it is.
- Minus P L by 6.
- This is the value that we are going to get.
- I hope all of you have understood how we can get the internal effects or what are the internal effects that is acting on a beam.
- How or what is the sign convention that should be used for shear force and bending movement diagram.
- Also, you have understood how do we get the distribution of the shear force and bending moment diagram between the zones or the points where there is no variation in the load, and how to get the value of the shear force and bending moment bending moment from the distribution.
- We'll be continuing in the next class.
- This was class number 53.
- In the next class, we will be solving a few more problems on how do we draw the bending moment and GFO diagram and how do we determine the shear force and bending movement diagram or shear force and bending movement values at a particular point.
- My name is Dr.
- I am from the civil engineering department of science and humanities PS University.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000146 · EV0000147 · EV0000148 · +1 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
UE25CV141A_Unit 3_Class 53_AVSummary transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/042_UE25CV141A_Unit_3_Class_53_AVSummary__fb2a14ea9c91ccb3.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE25CV141A_Unit 3_Class 53_AVSummary transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/042_UE25CV141A_Unit_3_Class_53_AVSummary__fb2a14ea9c91ccb3.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE25CV141A_Unit 3_Class 53_AVSummary transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/042_UE25CV141A_Unit_3_Class_53_AVSummary__fb2a14ea9c91ccb3.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9996; only the JSON segmentation supplies the displayed timestamps.
UE25CV141A_Unit 3_Class 53_AVSummary transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/042_UE25CV141A_Unit_3_Class_53_AVSummary__fb2a14ea9c91ccb3.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. 15 rendered pages in 033_Centroid__9f0e55503970de0e.slides.md have no readable `.slides.md` text; their original page images are included.
2. 2 rendered pages in 033_UE26ME142A_Unit_3_Class_33_Lecture_PPT__ce948fc3cd6a59bf.slides.md have no readable `.slides.md` text; their original page images are included.
3. 3 rendered pages in 039_UE24CV141A__Unit_3_Beams__9fc34bbd5ca57988.slides.md have no readable `.slides.md` text; their original page images are included.

4. 3 rendered pages in 039_UE25CV141A_Unit_3_Class_48_Lecture_PPT__4d28f283532e284e.slides.md have no readable `.slides.md` text; their original page images are included.
5. 3 rendered pages in 042_UE25CV141A_Unit_3_Class_53_Lecture_PPT__c20af5d61d14aaf0.slides.md have no readable `.slides.md` text; their original page images are included.
6. 5 rendered pages in 043_UE25CV141A_Unit_3_Class_54_Lecture_PPT__c7db1f404daea47d.slides.md have no readable `.slides.md` text; their original page images are included.
7. 6 rendered pages in 044_UE25CV141A_Unit_3_Class_55_Lecture_PPT__741ad989a632bc04.slides.md have no readable `.slides.md` text; their original page images are included.
8. 4 rendered pages in 044_UE26ME142A_Unit_3_Class_44.1_Lecture_PPT__068ee897e4f8117e.slides.md have no readable `.slides.md` text; their original page images are included.
9. No historical archive question page met the unit-match candidate threshold; all subject-level and ambiguous pages remain in datasets/question-archive.json.
10. 6 video-to-slide associations were not promoted to a match. Candidate rankings are listed in the transcript corpus and video-slide-matches.csv; the slide decks appear separately.
11. 1 transcript segments have low ASR confidence; they are flagged with timestamps in the transcript blocks and indexed in the corpus.
12. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
13. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
14. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
15. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00034	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0038. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0039 to s0058. Sidecar variants compare...
SRC00035	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0038. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0039 to s0058. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00036	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0038. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0039 to s0058. Sidecar variants compare...
SRC00037	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0038. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0039 to s0058. Sidecar variants compare...

C Revision and retrieval practice